Meeting of the Expert Group on waste related to the Batteries regulation

Methodology for the calculation and verification of the percentage of recycled content in batteries

15 October 2025

European Commission

Agenda

1. Welcome and adoption of the agenda

2. Preparatory work for the Commission Delegated Regulation on the methodology for the calculation and verification of the percentage of recycled content in batteries, and the format for the documentation, pursuant to Article 8 of Regulation (EU) 2023/1542

– Presentation of the working draft of the Annex to the Delegated Regulation

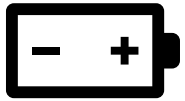
– Discussion on the working draft of the Annex to the Delegated Regulation

3. Any other business

Context and legal mandate

Article 8 of Regulation EU 2023/1542

- **Scope**:



Industrial / EV / SLI (+LMT from 2036)

Metals	in	recovered from
Cobalt	Active material	Battery manufacturing waste or post-consumer waste
Nickel		
Lithium		
Lead	The battery	waste

- **Targets**:

for **each battery model** per **year** and per **manufacturing plant**

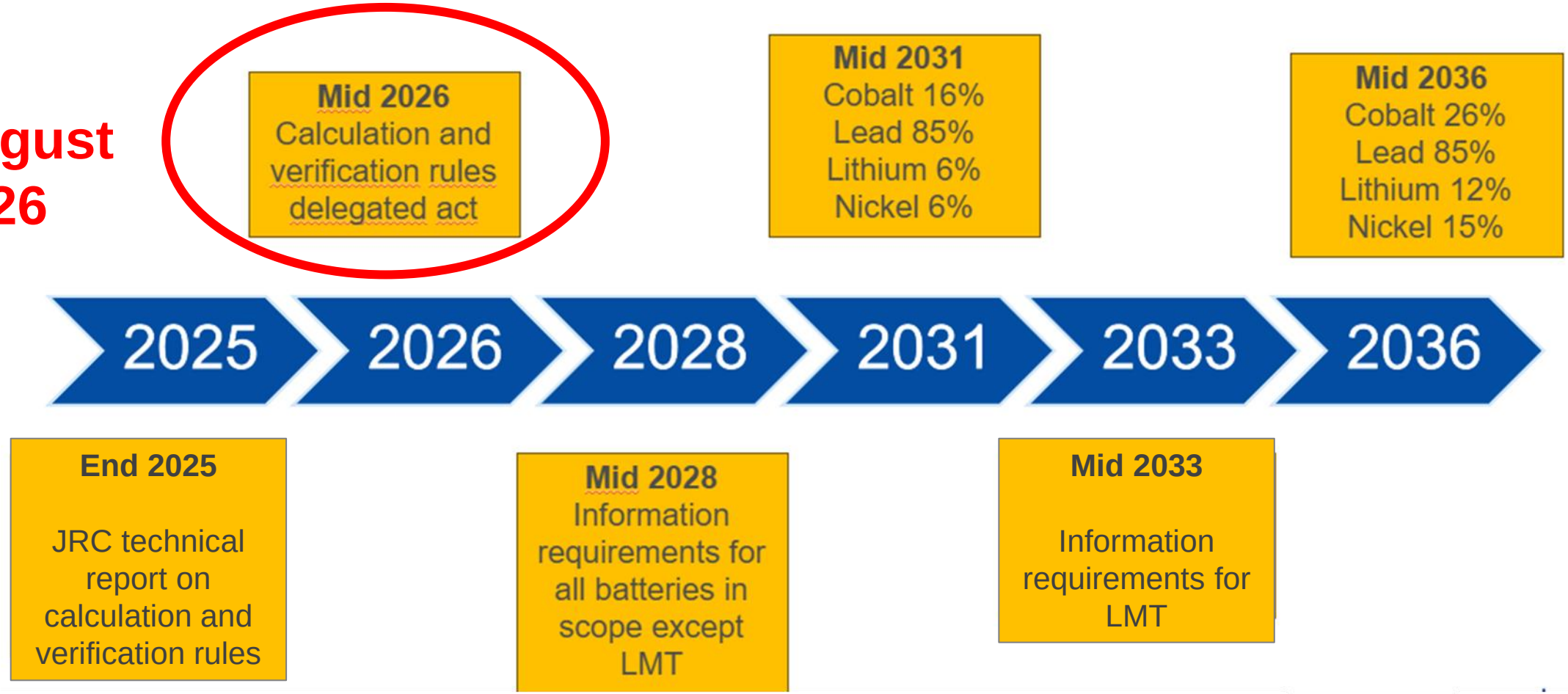
- **18.08.2031** – Minimum % of cobalt (16%), nickel (6%), lithium (6%) and lead (85%)
- **18.08.2036** - Minimum % of cobalt (26%), nickel (12%), lithium (15%) and lead (85%)

Legal mandate

- Article 8(1) 3rd subparagraph

“By 18 August 2026, the Commission shall adopt a delegated act in accordance with Article 89 to supplement this Regulation by establishing, for the batteries referred to in the first and second subparagraphs, the methodology for the calculation and verification of the percentage share of cobalt, lithium or nickel that is present in active materials and that has been recovered from battery manufacturing waste or post-consumer waste, and the percentage share of lead that is present in the battery and that has been recovered from waste, and the format for the documentation”

Timeline from the Batteries Regulation



JRC study

- JRC was mandated to formulate techno-scientific recommendations.
- Stakeholders consultations: 27 November 2024 & 3 April 2025
- Assessment of a range of possible options against different effects categories:

Business continuity and competitiveness

Resource efficiency and substitution of primary materials

Administrative burden

Straightforwardness of notified bodies to operate

Consumer's confidence in green claims and recycled content

Climate change mitigation and other environmental dimensions

https://rmis.jrc.ec.europa.eu/uploads/Draft_JRC_Report_ReCo_rules_Batt.pdf

Draft working document

Structure of the draft working document

- Working document – first draft of the annex of the future delegated act

- 3 main parts:

Definitions	Calculation rules (including chain of custody)	Verification rules & format for the documentation

- 2 options for the chain of custody model, embedded in the draft working document

I. Definitions

- Aims to define key terms for the future delegated act that are not already defined in Regulation (EU) 2023/1542

- Key definitions cover, inter alia:
  - Manufacturing plant
  - Input and output fractions
  - Chain of custody
  - Accounting period
  - Volume reconciliation

- All definition would apply for both options, except for **toll-treatment mechanism** only necessary for option 1

I. Definitions

(11) "Chain of custody" means a process by which inputs and outputs and associated information are transferred, monitored and controlled as they move through each step in the relevant supply chain;

(12) "Accounting period" means a timeframe defined for bookkeeping purposes, to reconcile input and output materials and to reallocate volumes of recycled material;

(13) "Volume reconciliation" means the reconciliation of inputs and outputs through the production process within the accounting period;

Calculation rules

II. Calculation rules

- Rules to calculate the recycled content in batteries meaning the proportion by mass of recycled to total cobalt, lithium or nickel in active materials of a battery or recycled to total lead in battery.

- 4 main sections:

Chain of custody and system description

Data collection and processing

Calculation for output fractions produced during the accounting period

Calculation of the documented recycled content

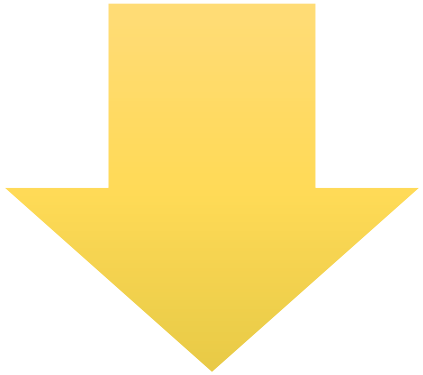
Calculation rules

Chain of custody model options

Choice of the options

In the absence of a technical method to measure directly in the batteries, recycled content has to be tracked througouht the supply chain through the implementation of a pre-defined chain of custody model

- Take into account the JRC analysis
- Finding the right balance

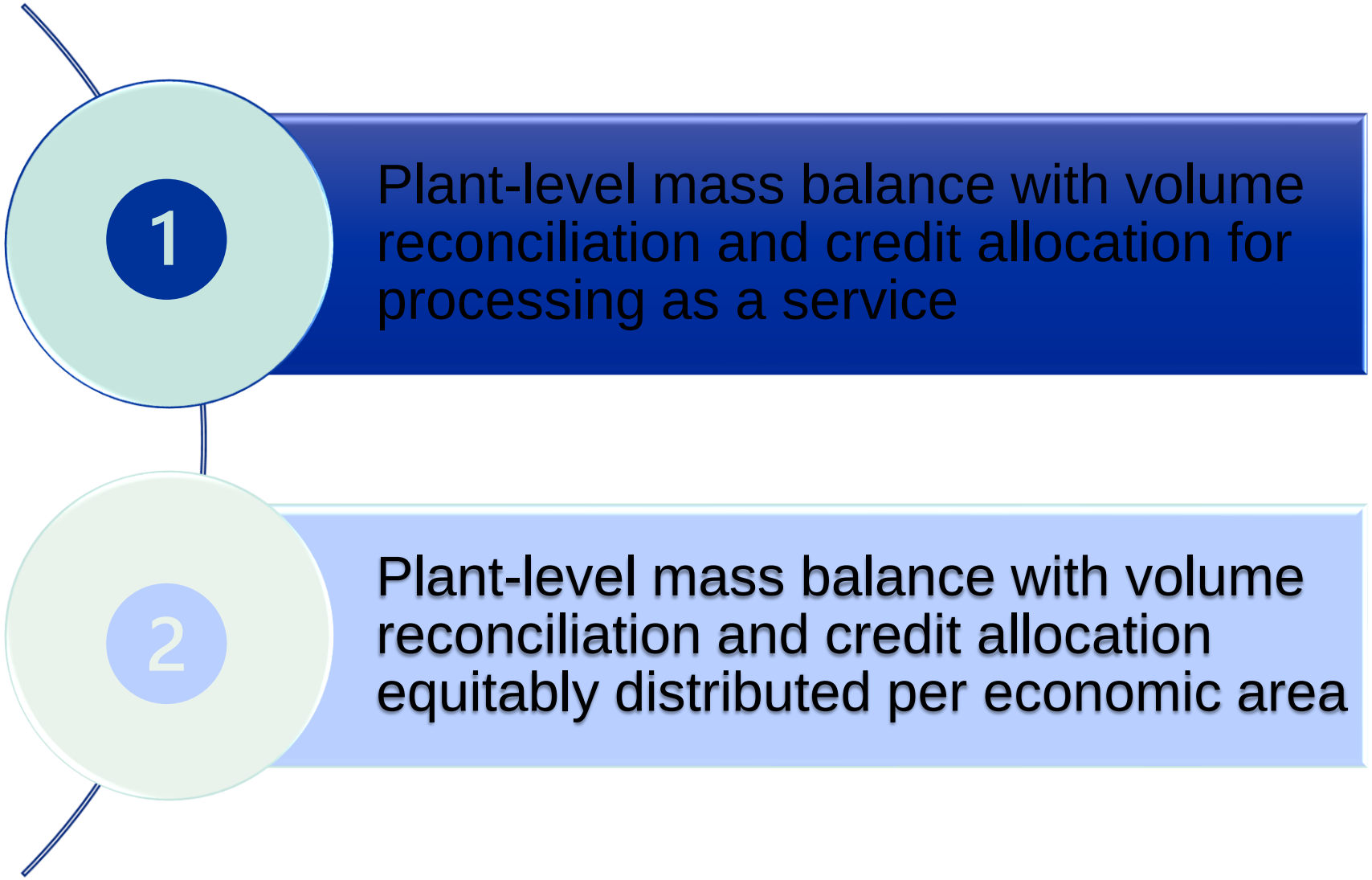


- Compliance with the Regulation
- Ensure consumers’trust
- Avoid circumvention of the Regulation



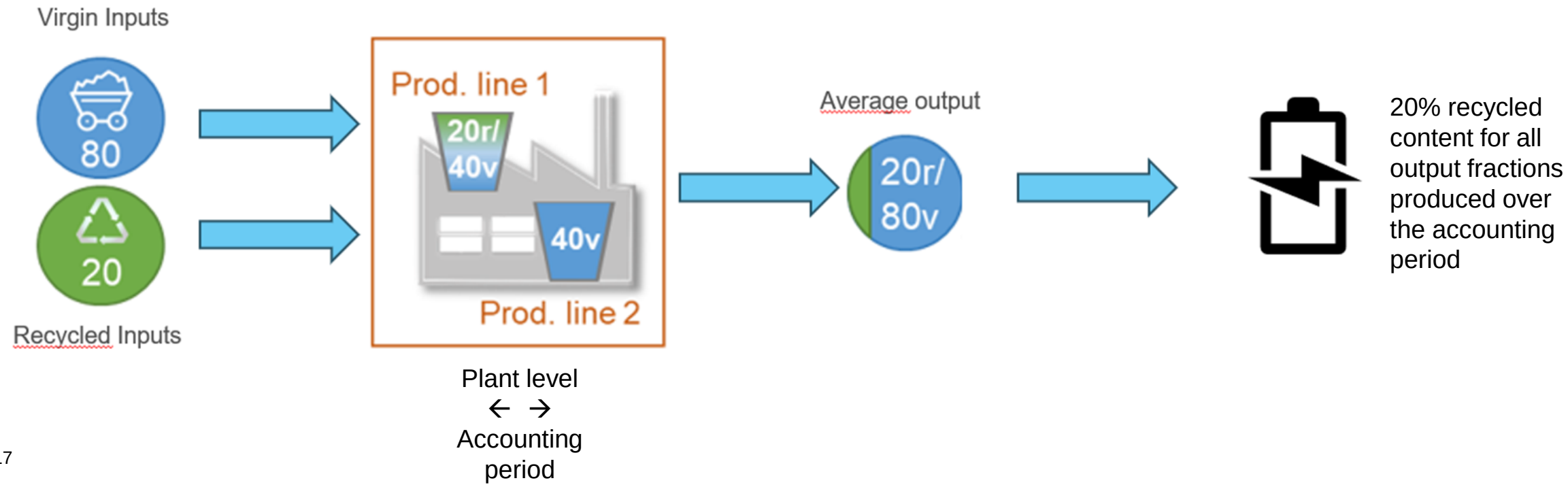
- Flexibility
- Limit financial and administrative burden
- Ensure level playing field

Description of the options



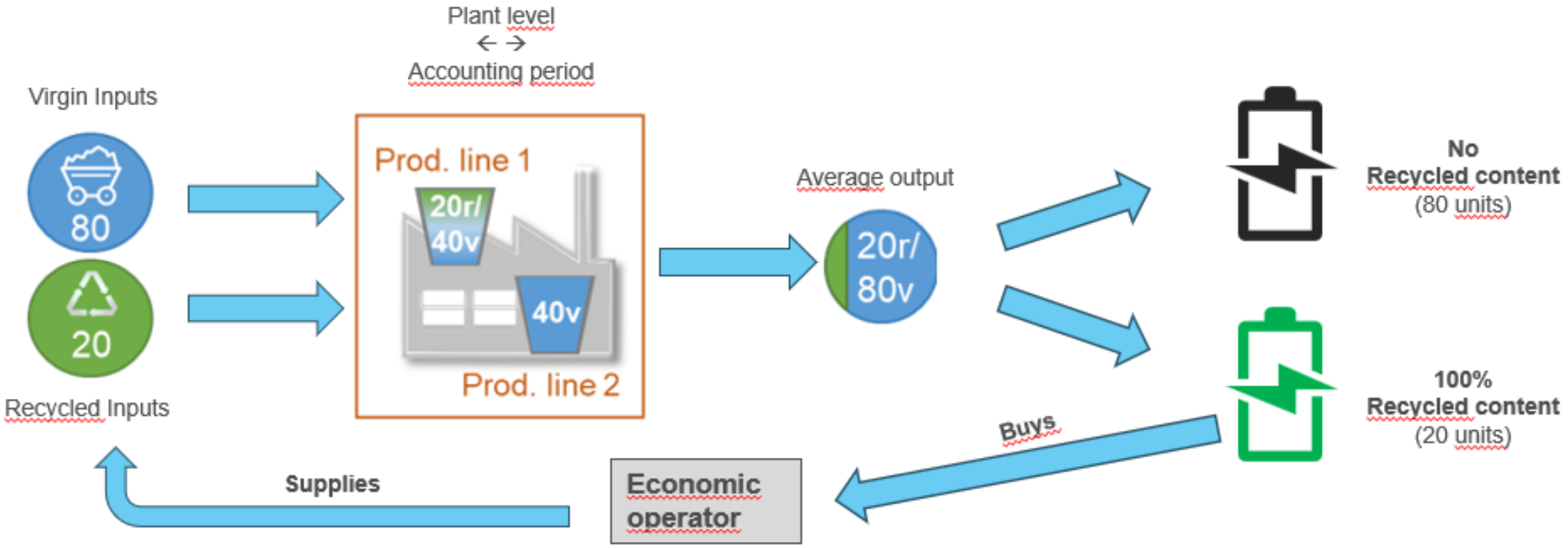
Option 1 - Plant-level mass balance with volume reconciliation and credit allocation for processing as a service

- Baseline: averaging the recycled content at plant level over the accounting period – output fractions have the same level of recycled content



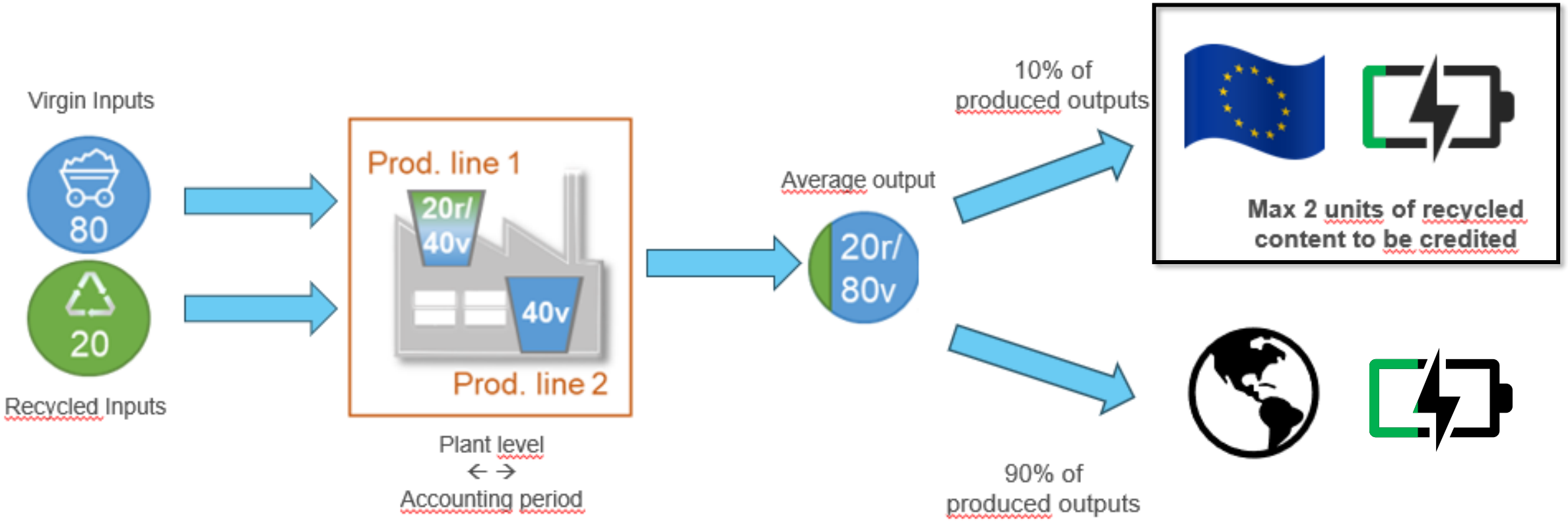
Option 1 - Plant-level mass balance with volume reconciliation and credit allocation for processing as a service

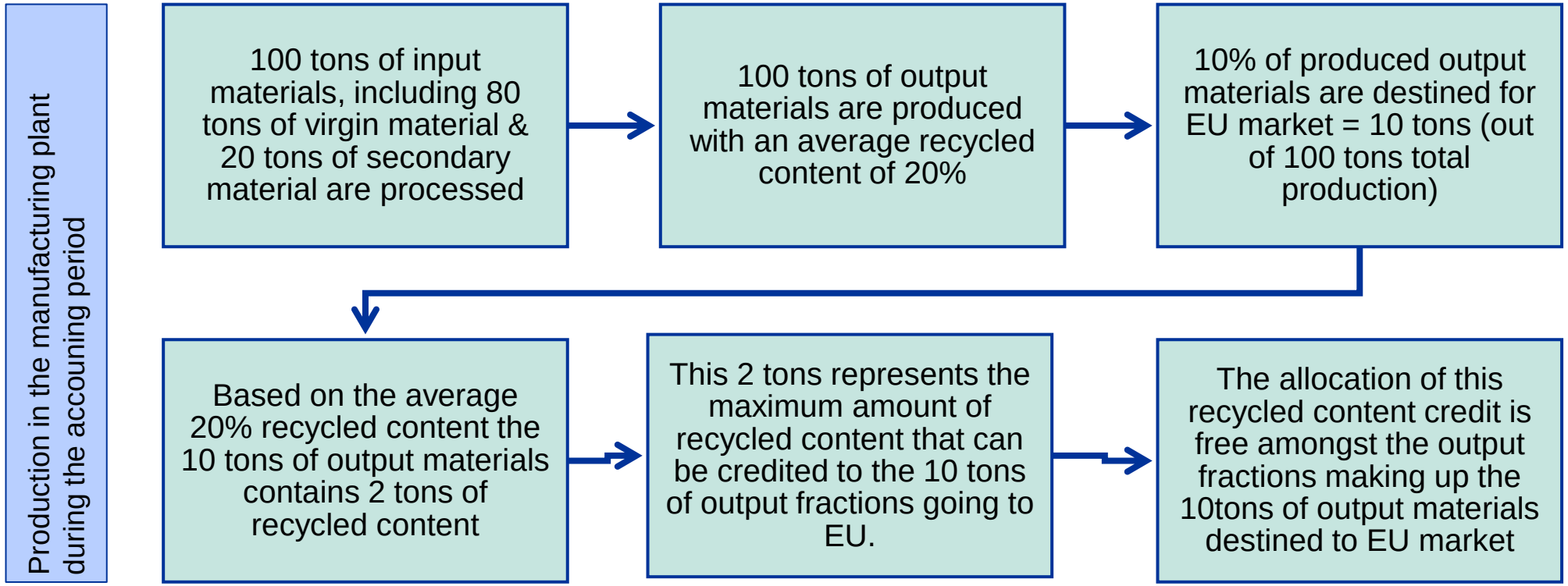
- In addition allocation of credits of recycled content is possible only when an actor supply the input fraction and receives the equivalent process material as the output fraction (toll-treatment mechanism).



Option 2 - Plant-level mass balance with volume reconciliation and credit allocation equitably distributed per economic area

- Free credit allocation, but maximum recycled content to be attributed to EU market is proportional to the share of outputs destined for the EU





Further explanation regarding the example in slide 19 to illustrate option 2 are provided in this slide.

Please note that all figures (mass of inputs, outputs, recycled content,…) are randomly chosen and that the process is simplified for illustration purposes. The rationale would remain valid for any other figures.

Examples of possible allocation of the 2 tons of recycled content considering, for example, that the 10 tons of output materials destined for the EU are composed of 4 different output fractions

Total mass of output fractions	**Output fraction 1 4 tons**	**Output fraction 2 2 tons**	**Output fraction 3 3 tons**	**Output fraction 4 1 ton**
Mass and % of recycled content Example 1	0.8 tons	0.4 tons	0.6 tons	0.2 tons
	20%	20%	20%	20%
Mass and % of recycled content Example 2	0.9 tons	0.35 tons	0.5 tons	0.25 tons
	22.5%	17.5%	16.7%	25%

Option 2 - Plant-level mass balance with volume reconciliation and credit allocation equitably distributed per economic area

- Outputs destined for the Union market contains the average amount of recycled content of all outputs produced at the manufacturing plant over the accounting period.

- Outputs fractions destined for the Union market can be allocated different levels of recycled content, as long as the maximum credit allocation (cap) is respected.

General principles

- **System boundary:** the manufacturing plant
- **Timeframe:** accounting period (up to 12months)
- **Volume reconciliation** during the accounting period (mass of secondary input fractions correspond to mass of recycled material in output fractions)
- **Allocation of recycled content:**
  - Option 1: proportional + credit if toll treatment mechanism
  - Option 2: cap but free allocation under the cap
- **Credit carryover:**
  - Option 1: No
  - Option 2: Yes for positive credits



Any questions?

- Please raise questions you may have regarding the understanding of option 1 and 2

Calculation rules

Data collection and equations

Data Collection

- Data about the content of recycled material in outputs to be transfered along the value chain
- Accounting periods should be defined in order to cover the 12 months documenting period.
- Calculation shall be done in a digital calculation tool

Calculation steps – option 1

Mass of recycled material of output fractions (conversion factor)

Recycled content of output fractions

Calculation of the documented recycled content

Accounting period

Calculation – Option 1

Recycled content during the accounting period outside the toll mechanism

Mass of recycled material in outputs produced during the accounting period outside the toll mechanism.

Based on conversion factor

$$ReCo(X)_{o,no\text{-}toll} = \frac{m(X)_{rec\ input\ no_toll}}{m(X)_{output,no-toll}} * 100$$

Total mass of outputs produced during the accounting period outside the toll mechanism.

Mass of recycled material in outputs produced during the accounting period under the toll mechanism.

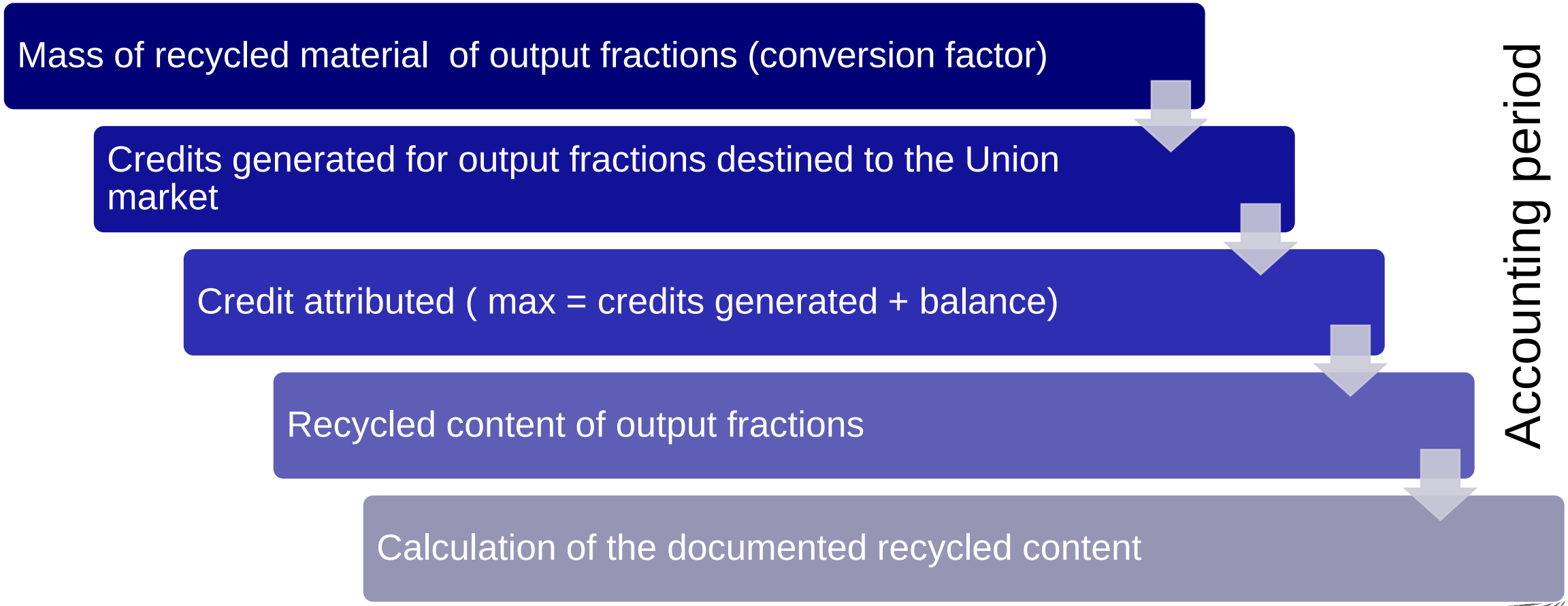
Based on conversion factor

$$ReCo(X)_{o,toll} = \frac{m(X)_{rec\ input\ toll}}{m(X)_{output,toll}} * 100$$

Recycled content during the accounting period under the toll mechanism

Total mass of outputs produced during the accounting period under the toll mechanism.

Calculation steps – option 2



Calculation – option 2

Credit generated during the accounting period

$$\mathbf{mc}(X_{EU}) = m(X)_{rec,output} * \frac{\sum m(X_{EU})_{tot,output}}{\sum m(X)_{tot,output}}$$

Total mass of outputs destined for the EU compared to total mass of outputs produced during the accounting period.

Mass of recycled material in outputs produced during the accounting period.

Based on conversion factor

Calculation – option 2

Mass of recycled material attributed to outputs destined for the Union market.

Based on Credit generated and credit balance from previous accoutning period

Recycled content during the accounting period

$$\mathbf{ReCo(X)_o} = \frac{mc(X)_{attributed}}{m(X)output} * 100$$

Total mass of outputs destined for the Union market produced during the accounting.



Discussion

- Please share your views and comments on the options suggested and the calculation rules



Coffee break

Verification rules

III. Verification rules

- Ensure that data are properly collected, processed and circulated along the supply chain by supply chain actors and recylced content properly documented over a 12 months period by battery manufacturers

- 2 main sections:

Format for the documentation

Responsibilities for notified bodies

Format for the documentation

- Annex VIII part B point 2

"2. ***Technical documentation***

The manufacturer shall draw up the technical documentation. The technical documentation shall make it possible to assess the battery's conformity with the relevant requirements referred to in point 1, and shall include an adequate analysis and assessment of the risks.

The technical documentation shall specify the applicable requirements and cover, as far as relevant for the assessment, the design, manufacture and operation of the battery. The technical documentation shall, where applicable, contain at least the following elements:

(…)

(i) a study supporting the recycled content share referred to in Article 8, containing the calculations made in accordance with the methodology set out in the delegated act adopted pursuant to Article 8(1), second subparagraph, and the evidence and information determining the input data for those calculations; and"

Format for the documentation – option 1

Table 1: general information about the manufacturer

Table 2: metal considered and accounting period

Table 3: information about outputs created during the accounting period

Table 4: recycled content of outputs produced during the accounting period

Table 5: documented recycled content

- Tables 2 to 4: repeated for all accouting periods composing the 12 months documentation period

Format for the documentation – Option 2

Table 1: general information about the manufacturer

Table 2: metal considered and accounting period

Table 3: information about outputs created during the accounting period

Table 4: credits attributed during the accounting period

Table 5: recycled content attributed during the accounting period

Table 6: documented recycled content

- Table 2 to 5: to be repeated for all accouting periods composing the 12 months documentation period

Responsibilities of the notified bodies

Verify digital calculation tool and application of methodology

Review inventory system and check absence of double counting

Check data accuracy, consistency and integrity

Review calculation and supporting information



Discussion

- Please share your views and comments on the verification rules

Next steps

Feedback by **Wednesday 12 November 2025**



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